

Where Does the Solenoid Burden Go?

A six-week feasibility study of merging-compression startup and NI-HTS centre-stack compatibility

PRIMARY QUESTION

How much central-solenoid burden can merging-compression genuinely remove, and where does that burden reappear?

The proposed first project connects two existing research streams: (1) sharp-D no-insulation high-temperature-superconducting toroidal-field windings, and (2) merging-compression and repetitive startup in spherical tokamaks. The initial task is not a complete reactor optimization. It is a bounded test of whether one representative startup transient creates a consequential magnet and recovery penalty.

Burden-transfer hypothesis

Reducing solenoid flux does not automatically remove startup cost. The burden may reappear as induced current, HTS hysteresis, structural eddy loss, TF field error, cryogenic recovery time, actuator power, failed formation shots, cyclic damage, or reduced plant duty.

Proposed first deliverable

- Reproduce one published charging/discharging response of the sharp-D NI-HTS winding.
- Apply one representative or sanitized CS/PF/merging startup waveform.
- Calculate the field seen by the winding, the field error, and the loss-channel decomposition.
- Issue a one-page verdict: signal, obstruction, indeterminate, or irrelevant.

Scope discipline

One primary proposal, with later branches activated only by the first result. No claim of novelty is made until the literature and group-internal work are checked.

A BOUNDED FIRST CALCULATION

Six-week feasibility sprint

- Week 1** Reproduce a published NI-HTS charging/discharging case and document model boundaries.
- Week 2** Define one startup cycle and compute the time-dependent field at the winding.
- Week 3** Add minimal vessel/casing/passive-current paths and check energy conservation.
- Week 4** Decompose cold energy by HTS hysteresis, structure/casing eddy, contact, coupling, and joints.
- Week 5** Sweep only the parameters that move the dominant channel and the useful plasma-field transfer.
- Week 6** Report the verdict, uncertainty, missing data, and whether a paper-scale model is earned.

Quantities to report

- Useful startup field at the plasma and unwanted TF-field perturbation.
- Cold joules per event, local temperature-rise proxy, and recovery time.
- Dominant loss channel and its sensitivity to waveform, contact, geometry, and passive structures.
- The central-solenoid flux or radial space actually saved at the same post-startup plasma target.

Kill tests and honest pivots

- Forcing at the winding is negligible: stop the magnet-coupling paper.
- HTS hysteresis dominates: study tape orientation, field angle, actuator placement, or slew.
- Casing/structure eddy loss dominates: study segmentation and field penetration.
- Contact-controlled response is material: test a small number of manufacturable resistance zones.
- Plasma startup uncertainty dominates: pivot to formation reliability and equivalent flux saved.

CONDITIONAL EXTENSIONS

What the first result could unlock

If the coupling is consequential

Compare CS-heavy, hybrid, and merge-heavy architectures at one fixed plasma mission. Parameterize startup allocation by $x_{CS} = \Phi_{CS}/\Phi_{\text{required}}$ and map feasibility in $(x_{CS}, \text{repetition rate})$. Report millimetres recovered, cold energy, field error, recovery, duty, and the active constraint.

If selective transmission appears possible

Test whether actuator placement, casing segmentation, tape orientation, winding topology/contact architecture, and waveform co-design can increase useful plasma field per unit cold-mass loss and TF-field distortion.

If a robust obstruction appears

Ask whether passivity or reciprocity imposes a quantitative limit on transmitting the startup band while suppressing loss and field error. This theory branch activates only after the numerical model displays the obstruction.

Inputs requested from Professor Tan

- The intended operating cycle: what repeats, what remains energized, and what sets dwell.
- One representative or normalized startup waveform and approximate actuator geometry.
- The metric that actually limits the NI-HTS winding or plasma scenario.
- Whether external mutual forcing can be added to the existing equivalent-circuit model.
- What work is already complete internally, and the publication/IP boundary.

Evidence base to reproduce first

Yin, Tan et al., *Fusion Engineering and Design* 220, 115338 (2025); Yin et al., *IEEE Transactions on Applied Superconductivity* 36, 4200305 (2026); Tan, Gao et al., IAEA FEC 2025 (SUNIST-2); Yang et al., *Superconductor Science and Technology* 38, 055008 (2025).